CONSTRUCTIBLE SUBSETS CUT DOWN TO A CLOSED SUBSET STAY CONSTRUCTIBLE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The criterion for constructibility through inverse images of retrocompact opens. Invoked at the start of the counted proof (source line 2881). Reduces the assertion to retrocompactness of a trace.
- The statement that a closed subset of a quasi-compact space is quasi-compact. Invoked at the end of the counted proof (source line 2888). Gives quasi-compactness of the trace.

Source reference. Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-inverse-images-constructibles: sources/stacks-topology/pinned-original-source.tex, source lines 2808–2808. Navigation pointer only: the counted text refers to the Stacks item lemma-inverse-images-constructibles, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a topological space. Let $Z \subset X$ be a closed subset such that $X \setminus Z$ is quasi-compact. Then for a constructible set $E \subset X$ the intersection $E \cap Z$ is constructible in Z.*

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Proof. Suppose that $V \subset X$ is retrocompact open in X. It suffices to show that $V \cap Z$ is retrocompact in Z by Lemma lemma-inverse-images-constructibles. To show this let $W \subset Z$ be open and quasi-compact. The subset $W' = W \cup (X \setminus Z)$ is quasi-compact, open, and $W = Z \cap W'$. Hence $V \cap Z \cap W = V \cap Z \cap W'$ is a closed subset of the quasi-compact open $V \cap W'$ as V is retrocompact in X. Thus $V \cap Z \cap W$ is quasi-compact by Lemma lemma-closed-in-quasi-compact. $\square$